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Neonatal diabetes is a disease that affects newly born children. The disease is caused by a change in the amino acid sequence of insulin.

This change prevents insulin binding to its receptor. Explain why this change prevents insulin binding to its receptor.

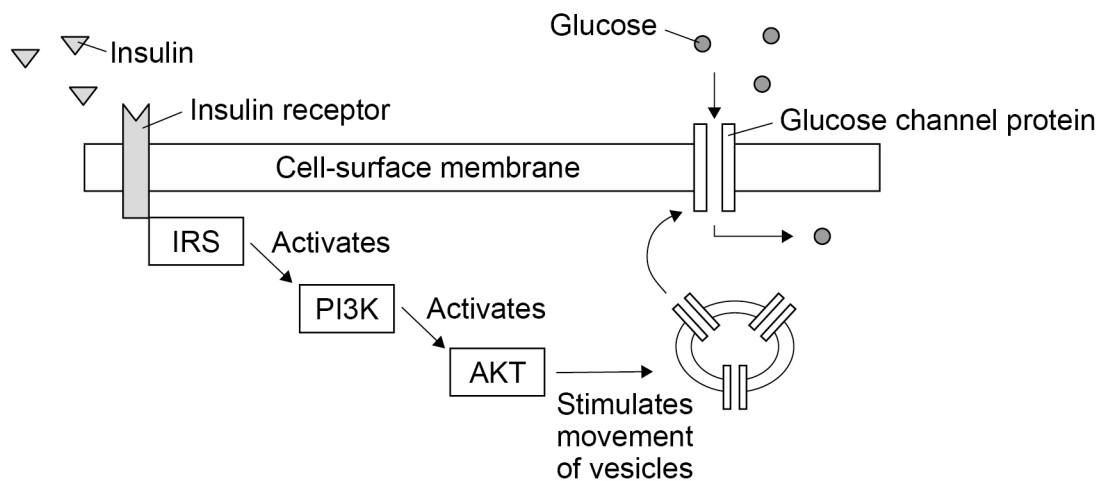
[2 marks]

Question 5 continues on the next page

Phosphoinositide 3-kinase (PI3K) is an enzyme in several metabolic processes.

Figure 5 shows the role of PI3K in the control of blood glucose concentration.

Figure 5



0 5 . 2 A decrease in the activity of PI3K can cause type II diabetes.

Use **Figure 5** to explain why.

[3 marks]

0 5 . 3

Using your knowledge of the kidney, explain why glucose is found in the urine of a person with untreated diabetes.

[3 marks]

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Turn over for the next question

Turn over ►

